
title: J · Risk Register description: Standalone bid-grade risk register — 18 tracked risks, likelihood × impact scoring, mitigation ownership, per-risk cost anchor, trigger events, residual score, and cross-references to the papers that describe each control.

J · Risk Register

!!! info “How this paper is used” Paper **B §15** carries a 12-item inline risk table for the compliance response. This paper is the **standalone register** an evaluator committee, a Program Steering Committee, or an internal QA gate expects — with 18 risks scored, owned, and cross-referenced.

A live copy of this register (Excel or ADO board) is created at M1 and updated weekly through M8;

J.1 Method

Scoring. Likelihood × Impact on a 1–5 scale each. Scores multiplied, banded per §J.2. **Owner.** Named role from [Paper F §F.3](#). Every risk has one accountable owner, not a shared responsibility. **Trigger.** The observable event that would cause the risk to fire. If you cannot state a trigger you cannot monitor the risk. **Control.** The specific paper section that describes the mitigation. Every control is traceable to a design decision, not to a vague “we will manage it.” **Residual score.** Score after the control is in place. Residual ≥ 12 requires escalation to the Program Steering Committee.

J.2 Risk-appetite bands

Score	Band	Action
1–5	Low	Accept; review quarterly
6–11	Medium	Mitigate under normal governance
12–19	High	Mitigate + weekly monitoring + PSC visibility
20–25	Critical	Board-level attention, contingency plan pre-approved

J.3 The register

J.3.1 Delivery risks

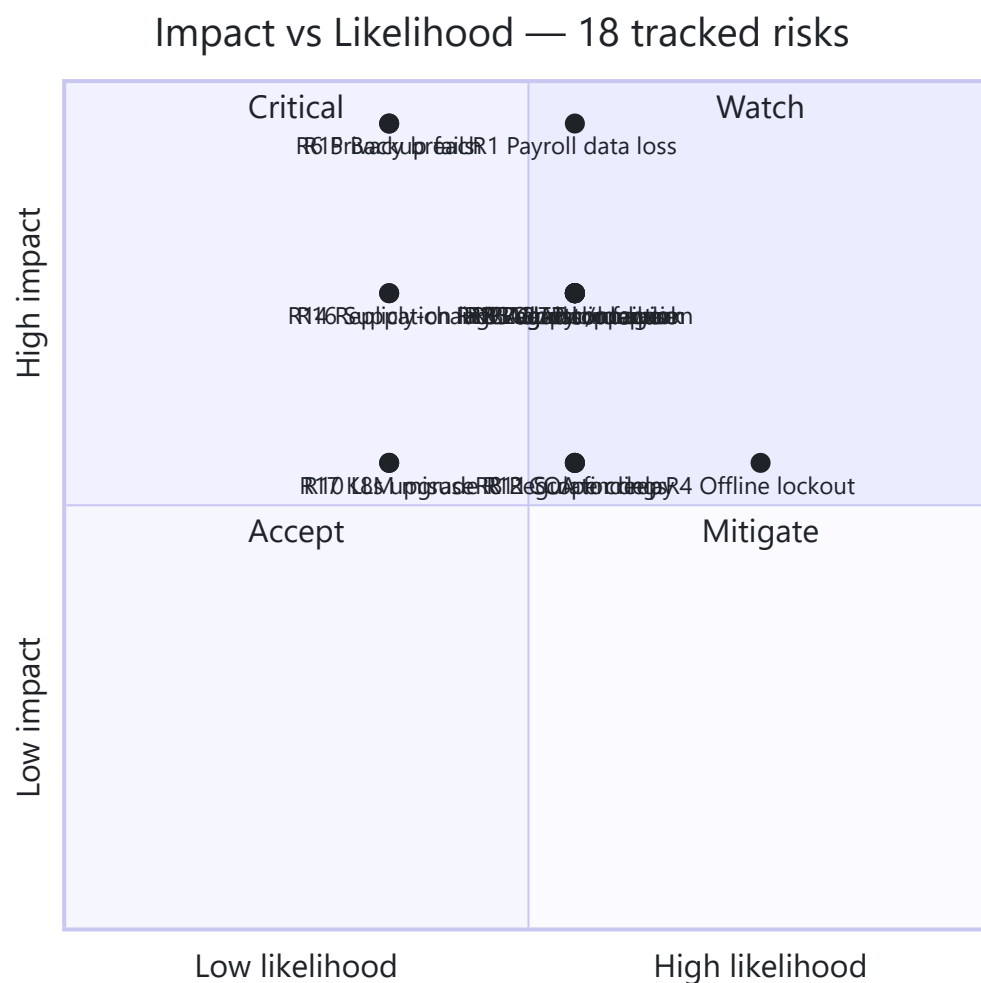
#	Risk	L	I	Score	Trigger	Control	Owner	Residual
R1	Payroll cutover data loss	3	5	15	Payroll parallel-run variance > 0.5% for ≥ 1 cycle	M6 parallel-run gate + statistical anomaly detector (D §4) · Paper H §H.7	Payroll Tech Lead	4
R2	Regulator-triggered rework (CSC / DBM / BIR issuance)	3	4	12	New DepEd Order or CSC memo published in build window	90-day adaptation SLA + versioned rule engine · Paper A §A.3.5	Solutions Architect	6
R3	Ghost / duplicate employees carried into new system	3	4	12	Dedup engine emits > 15 K ambiguous matches	PhilSys eKYC dedup + reconciliation engine (Paper H §H.5) + HR quarantine review	Migration Lead	4
R4	Low-connectivity schools locked out at go-live	4	3	12	Field survey shows > 15% of pilot SDOs unable to sync within 24 h	Offline PWA + CRDT sync (D §2) + SMS/USSD fallback (D §1)	Delivery Manager	4
R5	Vendor lock-in post-turnover	3	4	12	Prime contractor exits, DepEd cannot self-operate	10-yr source escrow + community edition (D §7) + full source turnover at M8	Program Manager	3
R6	Data privacy breach or unauthorised disclosure	2	5	10	SOC detects unauthorised query pattern, or NPC complaint filed	ISO 27001 / 27701 controls + hash-chain audit ledger (D §5) + Paper I breach protocol	Security Engineer + DPO	4
R7	UAT slippage compressing M7 into unusable window	3	4	12	UAT script pass-rate < 80% at M6 exit	Continuous UAT from M3 + env-parity discipline + M6 exit criteria enforced	QA Lead	6
R8	Integration partner delay (GSIS / Pag-IBIG / etc.)	3	3	9	Any regulator misses a sandbox milestone by > 30 d	Contract-first mocks + fallback batch mode + escalation to DICT	Solutions Architect	4
R9	Adoption failure at school level	3	4	12	ESS login rate < 40% at pilot SDOs after 30 d	Bilingual UI + SMS/USSD + trainer network + change-mgmt campaign (Paper F §F.8)	Change Management Lead	6

#	Risk	L	I	Score	Trigger	Control	Owner	Residual
R10	LLM misuse or hallucination in HR Copilot	2	3	6	User reports incorrect citation or unsafe output	Self-hosted retrieval-only model + guardrails + red-team eval (D §3)	Data Engineer	3
R11	Scope creep from module 4–11 discovery	3	3	9	> 3 new requirements accepted post-inception	Fixed inception window + change-control board + reference to §A.2	Delivery Manager	4
R12	Post-audit COA findings on disbursement	3	3	9	COA field audit within 90 d of go-live	Anomaly detector + transparency portal + hash-chain ledger + procurement-clean process	Program Manager	3

J.3.2 Technical risks

#	Risk	L	I	Score	Trigger	Control	Owner	Residual
R13	Peak-load payroll cycle triggers OLTP contention	3	4	12	Load test at M5 shows P95 > 3 s under 80 K concurrent	Sizing per Paper F §F.5.1 + read replica for reports + connection pooling + k6 load runs pre-M6	Solutions Architect	4
R14	PostgreSQL replication lag under sustained write load	2	4	8	Sync replica lag > 250 ms sustained	Streaming replication + monitored lag alerts + fallback to async replica for reports	Database Engineer	3
R15	Backup restoration fails at scale	2	5	10	Quarterly restore drill fails or exceeds RTO	Tested restore quarterly + immutable snapshots + off-site replicas	DevOps Lead	3
R16	Third-party dependency supply-chain compromise	2	4	8	Advisory or notarised CVE on a bundled dependency	SBOM per release + dependency signing + private registry + rapid-patch procedure	Security Engineer	4
R17	Kubernetes cluster upgrade breaks workload	2	3	6	Post-upgrade regression on non-prod	Progressive rollout policy + canary + version-N-1 support policy	DevOps Lead	3
R18	Legacy PIS data corruption discovered mid-migration	3	4	12	Reconciliation report flags > 1% variance in identity fields	Multi-source cross-check + HR quarantine review + wave-scoped rollback (Paper H §H.8)	Migration Lead	4

J.4 Heat-map view



J.5 Residual-score movement

Every risk in J.3 shows a **residual score** — the score *after* the control lands and is proven. The residual column is what a Steering Committee reads first.

Only **two risks retain a medium residual** (score ≥ 6) after mitigation:

- **R2 • Regulator rework** — driven by external issuances (CSC, DBM, BIR); we can compress but not eliminate.
- **R7 • UAT slippage** — driven by DepEd availability of UAT participants; we can mitigate through continuous UAT but the calendar constraint remains.
- **R9 • Adoption failure** — driven by school-level connectivity and staff readiness; mitigated by fallback channels but true residual is behavioural.

All others drop to Low. **No risk has a critical residual.**

J.6 Trigger monitoring — what gets watched, and where

Trigger	Data source	Cadence	Escalation
Payroll variance > 0.5%	Parallel-run reconciliation report	Every payroll cycle	Payroll TL → Delivery Mgr → PSC
Reconciliation variance on any regulator > 0.05%	Regulator remittance report	Weekly through M8, monthly post	Finance + DBA → Program Mgr
Dedup ambiguous matches > 15 K	Migration dashboard	Weekly	Migration Lead → HR Governance
Field survey offline-sync failure	Pilot SDO report	Post-pilot	Delivery Mgr → PSC
SOC anomaly on data-access pattern	SIEM + audit ledger	Real-time	SecEng → DPO
ESS login rate < 40% at SDO	Adoption dashboard	Weekly for 90 d post-go-live	Change Mgmt → Program Mgr
Load-test P95 > 3 s	k6 CI runs	Pre-M5, pre-M6, pre-M7	Solutions Arch → Delivery Mgr
Replication lag > 250 ms sustained	Prometheus alert	1-min	DBA on-call
Quarterly restore drill fail	Backup runbook	Quarterly	DevOps Lead → Program Mgr

J.7 Contingency budget draw-down

The 7% contingency line in [Paper F §F.9.1](#) is **PHP 17.5 M** for the build phase. Draw-down policy:

Trigger	Draw	Governance
R2 fires (regulator issuance mid-build)	Up to 3.0 M	Delivery Mgr sign-off
R7 fires (UAT slippage of ≤ 2 weeks)	Up to 4.0 M	Delivery Mgr + Program Mgr
R13 fires (perf regression requiring re-architecture)	Up to 5.0 M	Program Mgr + PSC
Any two Medium residual risks fire concurrently	Full 7%	PSC + client sign-off
Any risk moves from residual Low to Medium	Up to 2.0 M	Delivery Mgr sign-off
Reserved for unforeseeable	3.5 M	Held to end of build

J.8 Review cadence

- **Weekly (Delivery Mgr + Tech Leads)** — Trigger dashboard, any new risks, residual movements.
- **Monthly (Program Steering Committee)** — Full register, heat map, contingency ledger.
- **Milestone gates (M1, M2, M3, M4, M5, M6, M7, M8)** — Formal review; risks with residual ≥ 6 require written sign-off before payment claim.
- **On any breach or trigger fire** — Ad-hoc PSC meeting within 48 h.

J.9 Cross-references

- Delivery plan and cost model that funds mitigations → [Paper F](#)
- Migration-specific controls → [Paper H](#)
- Privacy-risk detail (R6) → [Paper I §I.8](#)
- Value-adds acting as controls → [Paper D §§ 2 · 3 · 4 · 5 · 7](#)
- Inline procurement-facing 12-risk table → [Paper B §15](#)